Advanced Techniques of Machine Translation

Assignment 2

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1. Romanization and False Friends: Experiment Design

a. Discuss the results for the experiment in detail. What does the table show?

➔ The results of the experiment highlights that the NMT model trained on the original Cyrillic data performed much better compared to the NMT model trained on the Romanized data, even though both models were trained in a similar fashion. This means that the Cyrillic model is better at aligning source and target subwords whereas the Romanized model struggled to differentiate between similar looking but semantically different meaning words. Very briefly put: romanizing Russian to use a similar latin script didn't help the model. It ended up introducing noise to the dataset that confused the model.

b. Did you expect this outcome?

➔ Yes, this outcome was expected because romanizing Russian increases the overlap on the surface but doesn't guarantee the similarity to transfer to the semantic meaning of the words. As someone who speaks 2 other non-latin based languages, this seems intuitive as "romanization" of the languages often yield words that may look similar to existing words but mean something else altogether.

c. What conclusions can you draw from the results?

➔ The results suggest that using the original Cyrillic script preserves important semantic meaning that help the model learn more accurate alignments between Russian and English. Romanization, while seemingly helpful for subword sharing, actually introduces ambiguity and harms translation quality. It confuses the model by making unrelated words look similar, which weakens its ability to distinguish "true friends" from "false friends." Overall, this shows that script differences carry meaningful information, and preserving them can be more beneficial for translation performance than forcing surface-level similarity.

d. Are there any other experiments you could run to further back up your conclusions?

➔ To further support the conclusion that romanization introduces ambiguity that the model struggles to resolve, I would experiment with adding a special prepended token (e.g., <ROMANIZED>, like the <CLS> tokens used for classification) to each source sentence in the romanized dataset. This token would explicitly tell the model that the input text has been romanized. If the model's performance improves with this additional cue, it would confirm that the loss in accuracy seen earlier comes from the model's inability to internally distinguish between scripts. In other words, the issue isn't just the use of Latin characters but the lack of explicit script context. This experiment would help verify whether providing that context can mitigate the confusion caused by false friends.

2. Tokenization and Interpreting Perplexity Scores: Experiment Design

a. Discuss the results for experiment 3. What does the table show about the performance of MorphGPT compared to GPT-2 Bade and GPT-2 Large?

➔ (Assumption: MorphGPT = Morph 200k) In experiment 3, MorphGPT outperformed GPT-2 Base in terms of perplexity and performed just slightly worse compared to GPT-2 Large. Considering that the only difference between MorphGPT and GPT-2 (base/Large) is just the tokenization, the conclusion is that MorphPiece tokenizer vastly helps a model improve it's perplexity compared to a BPE tokenizer.

b. Is the perplexity metric impacted by the different tokenization techniques? How might this affect the interpretation of the results in the table?
   - ➔ Yes, as mentioned in the previous answer, the tokenization technique does affect the perplexity metric of a given model. Morph 200k was able to compete with GPT-2 Large and beat GPT-2 base solely by using the MorphPiece tokenizer. The fact that a model with 200k params was able to perform so well compared to models with more than 100M+ params shows that tokenization techniques do help models perform better.

c. MorphPiece typically produces longer sequences than BPE. What is the impact of longer sequences on perplexity scores, everything else being equal? How does this relate to the results we see for MorphGPT?
   - ➔ Longer sequences generally make it harder for a model to achieve low perplexity because each sentence is represented by more tokens, which increases the total prediction steps and the chance of small errors accumulating. In principle, if two models have identical predictive ability at the word level, the one producing more tokens per sentence should have a slightly higher perplexity. However, in this case, MorphGPT still achieves lower perplexity despite having longer sequences. That suggests that the segmentation introduced by MorphPiece actually makes the token representations more meaningful and predictable, offsetting the disadvantage of longer input sequences. In other words, even though MorphPiece creates more tokens, those tokens align better with linguistic units, allowing the model to make more confident next-token predictions.

d. Based on these results and your understanding of tokenization methods, what further experiments would you propose to investigate the strengths and limitations of MorphPiece tokenization?
   - ➔ I would try to use this tokenization method in different languages and compare the results to standard BPE tokenization. Furthermore, I will also like to look at the training overhead for both MorphPiece and BPE tokenization to check if the trade-off would equal the gains in metrics like Perplexity, BLEU, etc.